

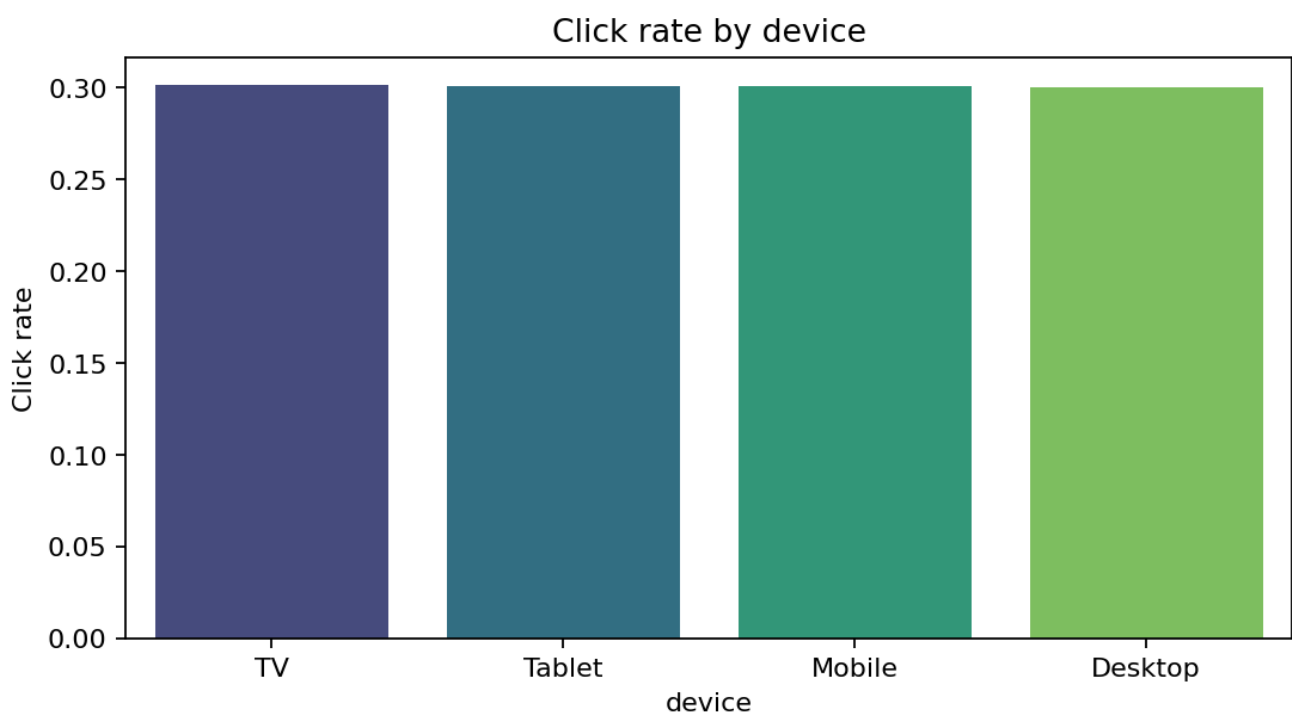
YouTube Recommendation Analysis

This report explores a large dataset of YouTube viewing activity, capturing how people interact with recommended videos. With nearly a million viewing records spanning tens of thousands of users and videos, the data offers a detailed look at what happens after a video is shown — whether people click, how much they watch, and whether they take deeper actions like liking, commenting, or subscribing. The headline finding is reassuring: the recommendation experience is remarkably steady and balanced across content types, devices, and times of day, with strong watching behavior but much rarer deep engagement.

The Big Picture

The dataset is substantial and well-suited for spotting reliable patterns. It contains 995,000 viewing records, 99,990 unique users, and 50,000 unique videos. Across all this activity, the recommendation funnel performs consistently:

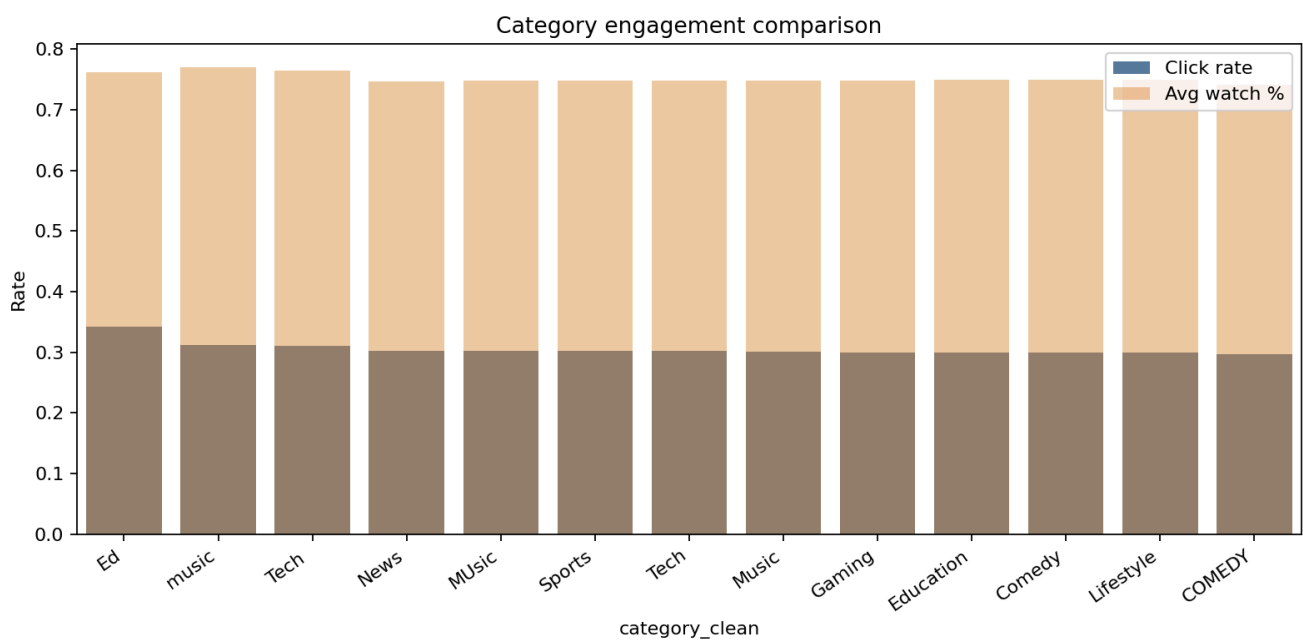
- About **30% of impressions lead to a click** — roughly 3 in 10.
- The **average watch percent is around 74.9%**, meaning people watch most of what they start.
- Nearly **half of all views (about 49.6%) reach full completion**.
- Only about **5% of views lead to a subscription**, showing that deep conversion is far harder to earn than a click or a view.



Engagement Is Balanced Across Categories

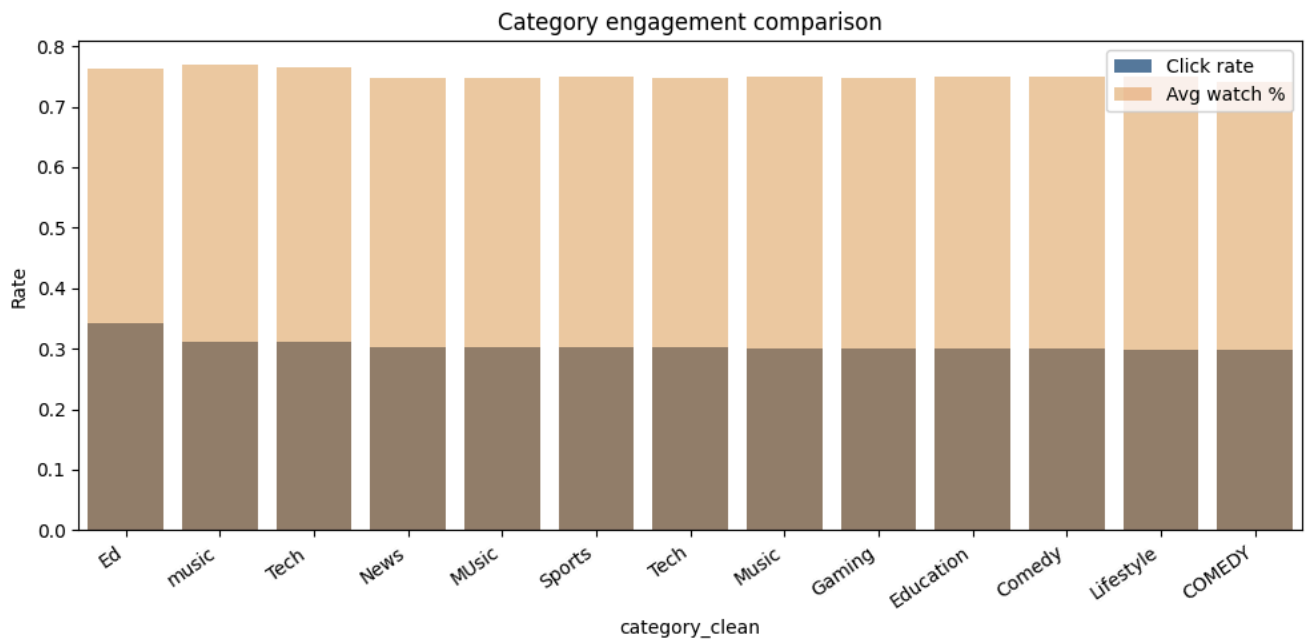
One of the clearest takeaways is how even the experience is across content types. Whether viewers are watching Comedy, News, Music, Tech, Sports, Gaming, Lifestyle, or Education, the click rates, watch depth, and subscription rates barely move. This suggests the recommendation system treats content types fairly, with no single category dominating.

Category	Rows	Click Rate	Watch Percent	Sub Rate
Comedy	124,467	0.300	0.749	0.050
News	123,920	0.303	0.747	0.051
Music	123,851	0.300	0.749	0.049
Tech	123,735	0.302	0.748	0.050
Sports	123,644	0.303	0.749	0.051
Gaming	123,595	0.300	0.749	0.051
Lifestyle	123,457	0.299	0.750	0.049
Education	123,331	0.300	0.749	0.049



Devices Behave the Same Way

The story is similar across devices. Mobile, Desktop, TV, and Tablet all see click rates near 30% and watch percentages around 75%. There is no meaningful drop-off tied to any one device, indicating a consistent experience no matter how people are watching.



Evening and Night Hold a Slight Edge

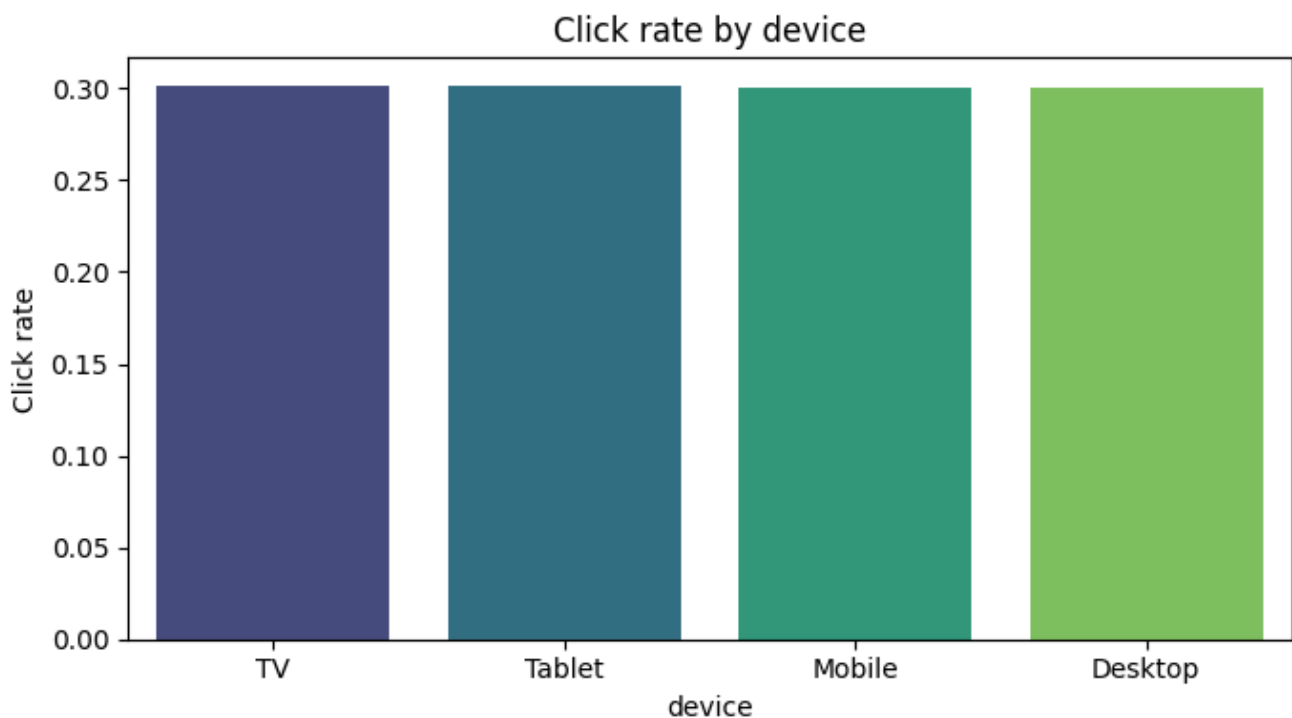
Time of day reveals a small but consistent pattern. Evening and night sessions edge out morning and afternoon across clicks, watch depth, and subscriptions. The differences are modest, but they point to when audiences are most responsive.

Time of Day	Rows	Click Rate	Watch Percent	Sub Rate
Morning	249,018	0.300	0.747	0.0498
Afternoon	248,989	0.300	0.749	0.0498
Evening	248,610	0.301	0.749	0.0502
Night	248,383	0.302	0.750	0.0503

Longer Videos Earn Time but Lose Completion

A clear relationship emerges between video length and how much of it gets watched. Longer videos accumulate more total watch time, but viewers are less likely to finish them — the average watch percent declines as duration grows. This trade-off is reflected in the data: video

duration and watch time move together (a correlation of about 0.61), while duration and completion percent pull apart (a correlation of about -0.44).



Bringing It Together

The analysis paints a picture of a steady, well-balanced viewing experience. Roughly 3 in 10 recommendations earn a click, viewers watch about three-quarters of what they start, and nearly half of all views run to completion — yet only about 1 in 20 leads to a subscription. Content categories and devices perform almost identically, evening and night sessions show a slight engagement advantage, and longer videos trade completion for total watch time. Together these findings highlight that while clicks and watching come easily and consistently, deeper engagement like subscribing remains the most challenging outcome to grow.